

GazeHTA: End-to-end Gaze Target Detection with Head-Target Association

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Problem

Who is looking at what?

Where is he looking?



Where is he looking?



Challenges

- Off-the-shelf head detectors
- Weak associations between heads and gaze targets

Contributions

- Scene feature from a pre-trained diffusion model
- Enhanced head priors by re-injecting head features
- Visual associations from connection maps

Method

Scene feature (F_{dec}) extraction

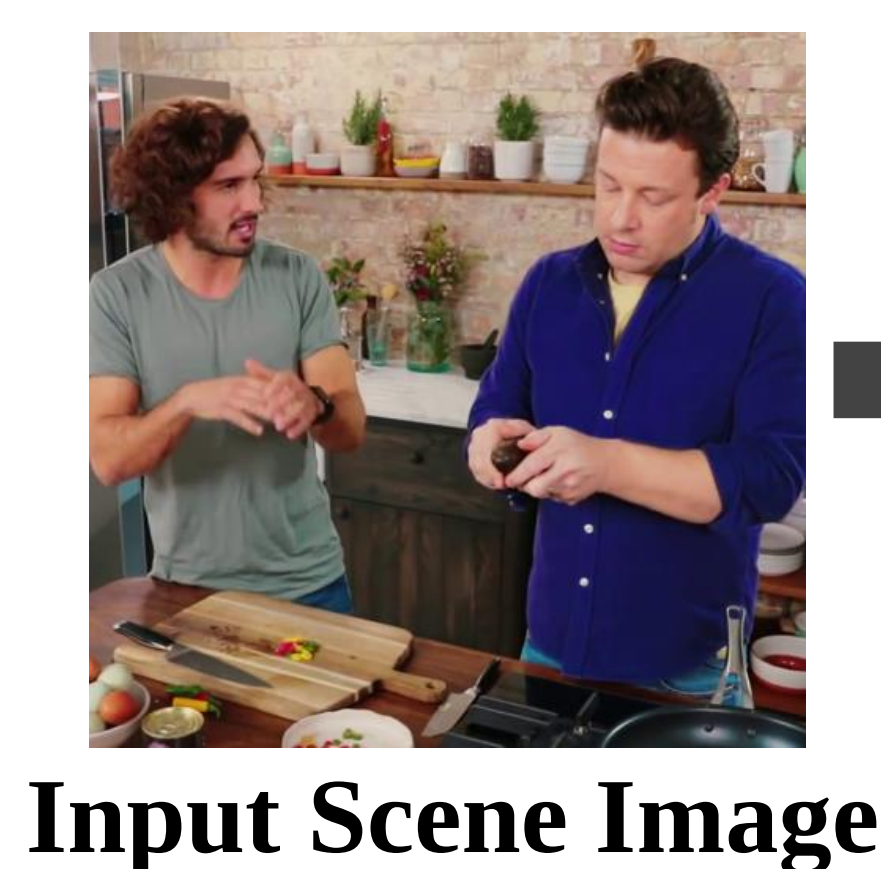
- GazeHTA directly predicts head-target pairs from images
- Leverage pre-trained diffusion model as scene feature extractor

Head feature (F_{head}) re-injection

- Learn F_{head} by supervision on head detection map (H_{det})
- Re-inject F_{head} back with F_{dec}

Connection maps

- Using the concatenated features (F_{prop}), GazeHTA predicts **head** and **target** pairs
- Auxiliary task to predict onnection map
- Heatmaps connecting head and target



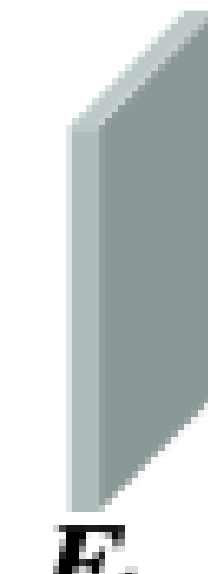
Pre-trained
Diffusion Model



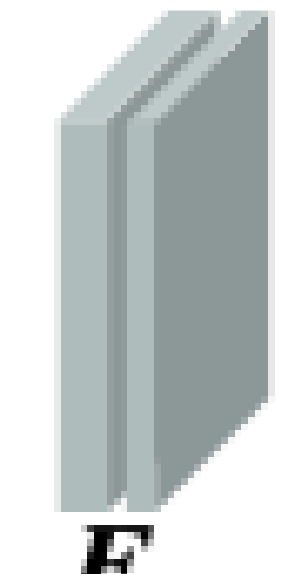
F_{dec}



H_{det}



F_{head}



F_{prop}



Head-target Instances

Experiments

- Datasets: GazeFollow, VideoAttentionTarget
- Baselines: Two-stream (up), end-to-end (mid)
- Results: GazeHTA outperforms baselines without off-the-shelf head detectors and constrained target objects

- Better combine head re-injection and connection maps
- Richer scene feature from pre-trained diffusion model

Method	GazeFollow				VideoAttentionTarget			
	AUC↑	Avg. Dist.↓	Min. Dist.↓	mAP↑	AUC↑	Dist.↓	AP↑	mAP↑
VideoAttention	0.921	0.137	0.077	-	0.860	0.134	0.853	-
DAM	0.922	0.124	0.067	-	0.905	0.108	0.896	-
ESCNet	0.928	0.122	-	-	0.885	0.120	0.869	-
Sharingan	0.944	0.113	0.057	-	-	0.107	0.891	-
HGTTR	0.917	0.133	0.069	0.547	0.893	0.137	0.821	0.514
Tonini et al.	0.922	0.072	0.033	0.573	0.923	0.102	0.944	0.607
GazeHTA	0.929	0.063	0.025	0.626	0.957	0.064	1.000	0.772

Backbone	C	F_{head}	GazeFollow			
			AUC↑	Avg. Dist.↓	Min. Dist.↓	mAP↑
Stable Diffusion [31]	$\times$	$\times$	0.910	0.084	0.042	0.500
	$\checkmark$	$\times$	0.931	0.064	0.027	0.601
	$\times$	$\checkmark$	0.928	0.068	0.029	0.567
	$\checkmark$	$\checkmark$	0.929	0.063	0.025	0.626
DETR [47]	$\checkmark$	$\checkmark$	0.928	0.064	0.028	0.607

Visualizations



Conclusion

- A further step toward fully end-to-end gaze target detection
- Better head-target association with head re-injection and connection maps
- State-of-the-art results on GazeFollow and VideoAttentionTarget